Seoyoung Oh

Problem-first Medical AI Researcher

Computer Vision · Reliable Evidence · Scientific Communication

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PROFESSIONAL PROFILE

Problem-first PhD researcher in Medical AI and Computer Vision, developing reliable and annotation-efficient methods for medical image analysis. Research spans weak supervision, generative and diffusion modeling, uncertainty, calibration, and explainable AI, complemented by industry experience in efficient computer vision deployment. Skilled at communicating complex evidence across technical, biomedical, and interdisciplinary audiences.

FRAME THE PROBLEM | DEVELOP THE SOLUTION | TEST THE EVIDENCE | COMMUNICATE THE MEANING

EDUCATION

Oct 2023 – Present

Ph.D. in Artificial Intelligence and Medical Image Analysis

ISEP • Laboratoire d'Imagerie Biomédicale • Sorbonne Université

- Supervisors: Prof. Jérémie Sublime and Prof. Véronique Marchand-Pauvert
- Supported by a Ph.D. grant from the French Ministry of Higher Education and Research
- Expected defense: September 2026

Sep 2021 – Sep 2023

Master in Artificial Intelligence, Computer Science

15.7/20

Université Paris-Saclay

Mar 2012 – Feb 2016

B.S. in Computer Science & Engineering

3.86/4.5

Seoul National University of Science & Technology

- Merit-based full scholarship: 2012 and 2015

PROFESSIONAL PROFILE

Oct 2023 – Present

Ph.D. Researcher

ISEP • Laboratoire d'Imagerie Biomédicale • Sorbonne Université

- Supervisors: Prof. Jérémie Sublime and Prof. Véronique Marchand-Pauvert
- Develop trustworthy and annotation-efficient methods for brain MRI analysis, spanning weak supervision, generative modeling, uncertainty, calibration, and explainable AI.
- Proposed Reliable Normal Projection with Healthy-Reference Calibration and evaluated it across BraTS, MSLesSeg, ADNI, and PULSE over five repeated runs.
- Achieved Pixel-AUPRC of 0.61 on BraTS and 0.48 on MSLesSeg, with a held-out healthy-test FPR of 0.047 at the calibrated operating point.
- Demonstrated 3.10× target-to-control ROI enrichment in ALS brainstem analysis, interpreted as anatomical plausibility rather than clinical validation.

PROFESSIONAL PROFILE

Mar 2023 – Aug 2023

Research Intern

ISEP • Laboratoire d'Imagerie Biomédicale • Sorbonne Université

- Supervisors: Prof. Jérémie Sublime and Prof. Véronique Marchand-Pauvert
- Investigated deep learning-based segmentation of the brainstem using neural networks pretrained on other anatomical regions.
- Compared transfer-learning strategies for a small and anatomically complex structure under limited annotation availability; work published at ISBI 2024.

Mar 2022 – Apr 2022

TER Project

Université Paris-Saclay • CNRS Laboratoire Interdisciplinaire des Sciences du Numérique

- Supervisor: Prof. Aurélie Névéol
- Investigated the use of personal names as proxies for social categories to evaluate potential bias in language models.

Dec 2017 – Mar 2020

Machine Learning Research Engineer, R&D

lululab Inc.

- Developed lightweight computer vision models for embedded Android devices without GPU acceleration.
- Worked on skin-problem detection, UV image analysis, semantic segmentation, and object localization.
- Improved skin-problem segmentation performance by 15.34% mIoU over DeepLabv3+.
- Developed a UV-reflective image analysis model using hierarchical dilated depthwise separable convolution, achieving 67.17% mIoU and 1.78-second inference without GPU acceleration.
- Optimized SSD/MobileNetV2-based object localization to 150 ms on an RK3399 Android board.

Jul 2015 – Jul 2017

Research Intern

Cryptography Information & Security Lab · Seoul National University of Science & Technology

- Supervisor: Prof. Changhoon Lee
- Studied blockchain-based approaches to improve trust in real-estate transactions.
- Researched cryptographic modes of operation suitable for data deduplication in cloud storage environments.
- Analyzed the security of 36 dual and 216 triple modes of operation against BOZ-PAD and ANSI X.923 padding-oracle attacks.
- Served as a teaching assistant for Data Structures coursework.

PUBLICATIONS & MANUSCRIPTS

2026 **Reliable Normal Projection with Healthy-Reference Calibration for Brain MRI Anomaly Localization**
European Conference on Computer Vision (ECCV), MedFM-Bench Workshop, 2026
Seoyoung Oh, Mélanie Pélégrini-Issac, Hélène Urien, Véronique Marchand-Pauvert, Jérémie Sublime

2026 **Multi-map Fusion for Weakly Supervised Disease Localization from Globally Assigned Diagnostic Labels in Brain MRI** [PDF]
IEEE International Symposium on Biomedical Imaging (ISBI), 2026
Seoyoung Oh, Mélanie Pélégrini-Issac, Hélène Urien, Véronique Marchand-Pauvert, Jérémie Sublime

2024 **Deep Neural Networks Comparison for MRI Segmentation of the Brainstem** [PDF]
IEEE International Symposium on Biomedical Imaging (ISBI), 2024
Seoyoung Oh, Mélanie Pélégrini-Issac, Hélène Urien, Véronique Marchand-Pauvert, Jérémie Sublime

2019 **RethNet: Object-by-Object Learning for Detecting Facial Skin Problems** [PDF]
IEEE International Conference on Computer Vision (ICCV), Visual Recognition for Medical Images Workshop, 2019
Shohrukh Bekmirzaev, Seoyoung Oh, Sangwook Yoo

2017 **Block Chain Applied Technology to Improve Reliability of Real Estate Market** [PDF]
The Journal of Society for e-Business Studies, 22(1), pp. 51–64, 2017
Seoyoung Oh, Changhoon Lee

TECHNICAL & SCIENTIFIC EXPERTISE

CORE EXPERTISE

- Computer Vision
- Medical Image Analysis
- Generative Modeling
- Diffusion Models
- Representation Learning

RESEARCH APPROACHES

- Weakly Supervised Learning
- Anomaly Detection
- Segmentation & Localization
- Annotation-Scarce Learning
- Healthy-Only Modeling

RELIABILITY & INTERPRETATION

- Trustworthy AI
- Explainable AI
- Uncertainty & Calibration
- Robustness & Domain Shift
- Evidence Evaluation

RESEARCH & ENGINEERING

- Python
- PyTorch & TensorFlow
- Experimental Design
- Reproducible ML Pipelines
- Deployment Optimization

LANGUAGES

Korean — Native · English — Advanced · French — Intermediate

SCIENTIFIC COMMUNICATION & CROSS-FUNCTIONAL STRENGTHS

SCIENTIFIC COMMUNICATION

- Scientific Writing
- Oral Presentation
- Visual Communication
- Audience Adaptation
- Interdisciplinary Collaboration

CROSS-FUNCTIONAL STRENGTHS

- Problem-First Reasoning
- Critical Evidence Interpretation
- Technical-to-Medical Translation
- Communication of Strengths & Limitations
- Real-World Implementation under Constraints

RESEARCH INTERESTS

Medical AI · Computer Vision · Weak Supervision · Diffusion Models · Reliable Evidence · Annotation Scarcity · Uncertainty & Calibration · Explainable AI · Scientific Communication